A First Course in Abstract Algebra

Section 48: Automorphisms of Fields

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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48 Automorphisms of Fields

Exercise 48.1: Conjugates of a square root

The minimal polynomial over $\mathbb{Q}$ is $x^2 - 2$, irreducible by Eisenstein at 2. Its two zeros are $\sqrt{2}, -\sqrt{2}$, so these are exactly the conjugates.

Exercise 48.2: Changing the base field

Since $\sqrt{2} \in \mathbb{R}$, its minimal polynomial over $\mathbb{R}$ is $x - \sqrt{2}$. Its only conjugate over $\mathbb{R}$ is itself. Conjugacy always depends on the base field.

Exercise 48.3: A translated square root

Subtracting 3 gives $(x-3)^2 - 2 = x^2 - 6x + 7$. This is irreducible over $\mathbb{Q}$ because a rational zero would make $\sqrt{2}$ rational. Its roots, hence all conjugates, are $3 + \sqrt{2}$ and $3 - \sqrt{2}$.

Exercise 48.4: A difference of square roots

Independent changes of signs of $\sqrt{2}$ and $\sqrt{3}$ give $\pm\sqrt{2} \pm \sqrt{3}$. They are four distinct conjugates of $\sqrt{2} - \sqrt{3}$. Squaring twice gives $x^4 - 10x^2 + 1 = 0$. To see the degree is 4, put $\alpha = \sqrt{2} - \sqrt{3}$. Since $\alpha^{-1} = -(\sqrt{2} + \sqrt{3})$, both square roots lie in $\mathbb{Q}(\alpha)$. Thus this field is the degree-four field $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, and the list is complete.

Exercise 48.5: A mixed real and imaginary generator

The four sign choices $\pm\sqrt{2} \pm i$ are the images under the four automorphisms of $\mathbb{Q}(\sqrt{2}, i)$. They are distinct. The field has degree 4 because i is not in the real field $\mathbb{Q}(\sqrt{2})$. Thus the minimal polynomial is $x^4 - 2x^2 + 9$, obtained by multiplying the four linear factors, and the four listed numbers are all conjugates.

Exercise 48.6: Conjugacy over the real field

Over $\mathbb{R}$, the minimal polynomial is $(x - \sqrt{2})^2 + 1$, which has no real zero. The conjugates are therefore $\sqrt{2} + i$ and $\sqrt{2} - i$. The base field fixes $\sqrt{2}$, so its sign cannot change.

Exercise 48.7: A nested square root over the rationals

Let $\alpha = \sqrt{1 + \sqrt{2}}$. Its square gives $\sqrt{2} = \alpha^2 - 1$, so $\mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\alpha)$. The element $1 + \sqrt{2}$ is not a square in $\mathbb{Q}(\sqrt{2})$: its norm to $\mathbb{Q}$ is -1, whereas the norm of a square is a rational square. Hence $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$. Eliminating the inner root gives $x^4 - 2x^2 - 1$. Its four zeros are $\pm\sqrt{1 + \sqrt{2}}$ and $\pm\sqrt{1 - \sqrt{2}}$, with the latter square root taken in $\mathbb{C}$. These are all conjugates.

Exercise 48.8: A nested square root over a quadratic base

Over $\mathbb{Q}(\sqrt{2})$, the polynomial is $x^2 - (1 + \sqrt{2})$. The norm argument in Exercise 48.7 shows it is irreducible. Thus only the two outer signs can change: the conjugates are $\sqrt{1 + \sqrt{2}}$ and $-\sqrt{1 + \sqrt{2}}$.

Exercise 48.9: A radical fixed by a sign change

The automorphism τ_2 changes only the sign of $\sqrt{2}$. Therefore $\tau_2(\sqrt{3}) = \sqrt{3}$.

Exercise 48.10: Applying a sign change to a sum

An automorphism preserves addition and fixes the other generating radicals. Thus $\tau_2(\sqrt{2} + \sqrt{5}) = -\sqrt{2} + \sqrt{5}$.

Exercise 48.11: Composing two sign changes

Apply the rightmost map first. The map τ_2 sends $\sqrt{2} + 3\sqrt{5}$ to $-\sqrt{2} + 3\sqrt{5}$; τ_3 fixes both radicals appearing there. Hence $(\tau_3\tau_2)(\sqrt{2} + 3\sqrt{5}) = -\sqrt{2} + 3\sqrt{5}$.

Exercise 48.12: An automorphism of a quotient

The map $\tau_5\tau_3$ flips $\sqrt{5}$ and $\sqrt{3}$ and fixes $\sqrt{2}$. Because field automorphisms preserve division,

$$(\tau_5\tau_3)\left(\frac{\sqrt{2} - 3\sqrt{5}}{2\sqrt{3} - \sqrt{2}}\right) = \frac{\sqrt{2} + 3\sqrt{5}}{-2\sqrt{3} - \sqrt{2}}.$$

The denominator remains nonzero because an injective map cannot send a nonzero element to zero.

Exercise 48.13: Simplifying a power of an automorphism

Since τ_5^2 is the identity and $\sqrt{45} = 3\sqrt{5}$, the remaining $\tau_3\tau_2$ flips only the $\sqrt{2}$ term. The result is $-\sqrt{2} + \sqrt{45}$.

Exercise 48.14: A nested composition

First $(\tau_2\tau_5)(\sqrt{30}) = \sqrt{30}$ because two radical signs change. Applying τ_5 to the bracket gives $\sqrt{2} - \sqrt{3} - \sqrt{30}$. Finally τ_3 changes the signs of the last two radical terms. The answer is $\sqrt{2} + \sqrt{3} + \sqrt{30}$.

Exercise 48.15: Fixed fields in a biquadratic extension

Write an element uniquely as $a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}$ with rational coefficients. In the source notation σ_1 flips $\sqrt{2}$, σ_2 flips $\sqrt{3}$, and σ_3 flips both.

(a) Fixing both σ_1 and σ_3 forces $b = c = d = 0$, so the fixed field is $\mathbb{Q}$.

(b) Fixing σ_3 forces $b = c = 0$, leaving $\mathbb{Q}(\sqrt{6})$.

(c) Fixing both σ_2 and σ_3 again forces $b = c = d = 0$, giving $\mathbb{Q}$.

Exercise 48.16: The fixed field of one radical flip

Use the basis $1, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{10}, \sqrt{15}, \sqrt{30}$ of E. The map τ_3 negates exactly those basis terms containing $\sqrt{3}$. Their coefficients must vanish in a fixed element. The surviving terms span $\mathbb{Q}(\sqrt{2}, \sqrt{5})$, which is the fixed field.

Exercise 48.17: The square of a radical flip

The map τ_3^2 is the identity, so every element is fixed. Its fixed field is $E = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$.

Exercise 48.18: Fixing two separate sign changes

An element fixed by both τ_2 and τ_3 has no basis term involving $\sqrt{2}$ or $\sqrt{3}$. Only $1, \sqrt{5}$ survive, so the fixed field is $\mathbb{Q}(\sqrt{5})$.

Exercise 48.19: Fixing two simultaneous sign changes

The map $\tau_5\tau_2$ negates $\sqrt{2}, \sqrt{5}$ and fixes $\sqrt{3}$. A basis monomial survives exactly when it contains both or neither of $\sqrt{2}, \sqrt{5}$. The surviving basis is $1, \sqrt{3}, \sqrt{10}, \sqrt{30}$, so the fixed field is $\mathbb{Q}(\sqrt{3}, \sqrt{10})$.

Exercise 48.20: Fixing all three simultaneous signs

A basis monomial is fixed by $\tau_5\tau_3\tau_2$ exactly when it contains an even number of the three radicals. Thus the fixed field has basis $1, \sqrt{6}, \sqrt{10}, \sqrt{15}$ and equals $\mathbb{Q}(\sqrt{6}, \sqrt{10})$, since $\sqrt{6}\sqrt{10} = 2\sqrt{15}$.

Exercise 48.21: Fixing each of the three generators

Every nonconstant basis monomial contains at least one generating radical and is negated by its sign flip. Requiring invariance under all three flips leaves just rational multiples of 1. The fixed field is $\mathbb{Q}$.

Exercise 48.22: The eight-element sign-change group

(a) Each τ_j is nonidentity, since it negates a nonzero radical, and its square fixes all generators. It therefore has order 2.

(b) Put $a = \tau_2, b = \tau_3, c = \tau_5$. They commute because independent sign changes commute on each

generator. The elements are $1, a, b, c, ab, ac, bc, abc$, with table

	1	a	b	c	ab	ac	bc	abc
1	1	a	b	c	ab	ac	bc	abc
a	a	1	ab	ac	b	c	abc	bc
b	b	ab	1	bc	a	abc	c	ac
c	c	ac	bc	1	abc	a	b	ab
ab	ab	b	a	abc	1	bc	ac	c
ac	ac	c	abc	a	bc	1	ab	b
bc	bc	abc	c	b	ac	ab	1	a
abc	abc	bc	ac	ab	c	b	a	1

The rule is to cancel repeated letters, since $a^2 = b^2 = c^2 = 1$.

(c) Any rational automorphism sends each $\sqrt{j}$ to $\pm\sqrt{j}$ because it preserves its minimal polynomial. There are at most $2^3 = 8$ choices. The eight maps above realize every choice, so they form the full group, isomorphic to $\mathbb{Z}_2^3$.

Exercise 48.23: Correcting the conjugate criterion

Replace "the same polynomial" by "the same irreducible polynomial over F." Equivalently, α, β have the same monic minimal polynomial over F. Merely being roots of one polynomial is insufficient: 0 and 1 are both roots of $x(x-1)$, but have different minimal polynomials.

Exercise 48.24: The kernel criterion for conjugacy

The statement is correct. The evaluation kernel at an algebraic element is the principal ideal generated by its monic minimal polynomial. Equality of the kernels is equivalent to equality of their monic generators, hence to conjugacy over F.

Exercise 48.25: A shifted generator gives the same automorphism

(a) The other conjugate of $\alpha = 3 + \sqrt{2}$ is $\beta = 3 - \sqrt{2}$, since both satisfy $x^2 - 6x + 7$.

(b) Both conjugation automorphisms fix $\mathbb{Q}$ and send $\sqrt{2} = \alpha - 3$ to $-\sqrt{2} = \beta - 3$. Since $\sqrt{2}$ generates the field, they are exactly the same map; only the chosen generator used to describe it differs.

Exercise 48.26: Frobenius on four elements

Write the field as $\mathbb{Z}_2(\alpha)$ with $\alpha^2 + \alpha + 1 = 0$, so $\alpha^2 = \alpha + 1$. Squaring gives

$$0 \mapsto 0, \quad 1 \mapsto 1, \quad \alpha \mapsto \alpha + 1, \quad \alpha + 1 \mapsto \alpha.$$

Thus precisely 0 and 1 are fixed; the fixed field is $\mathbb{Z}_2$.

Exercise 48.27: Frobenius on nine elements

Here $\alpha^2 = -1 = 2$ in characteristic 3. For $a, b \in \mathbb{Z}_3$, cubing gives $(a + b\alpha)^3 = a - b\alpha$. Thus 0,1,2 are fixed; the other six elements are exchanged in the pairs

$$\alpha \leftrightarrow 2\alpha, \quad 1 + \alpha \leftrightarrow 1 + 2\alpha, \quad 2 + \alpha \leftrightarrow 2 + 2\alpha.$$

The fixed field is $\mathbb{Z}_3$.

Exercise 48.28: Frobenius on an infinite field

Let $F = \mathbb{Z}_p(t)$ with t transcendental. The image of $r(t) \mapsto r(t)^p$ is $\mathbb{Z}_p(t^p)$. It does not contain t: if $(P(t)/Q(t))^p = t$ with P, Q nonzero polynomials, comparing degrees in $P(t)^p = tQ(t)^p$ gives $p \deg P = 1 + p \deg Q$, impossible modulo p. Frobenius remains injective, but need not be onto.

Exercise 48.29: Ten statements about field automorphisms

(a) False. An automorphism fixes 0 and 1 and cannot map one to the other.

(b) False. $\mathbb{Q}(0) = \mathbb{Q}$ and $\mathbb{Q}(\sqrt{2})$ are not isomorphic: any isomorphism fixes $\mathbb{Q}$, and their rational degrees differ.

(c) True, by the conjugation isomorphism theorem.

(d) True. It fixes 1, hence integers and their quotients, giving the prime field.

(e) False. The identity automorphism of a finite field fixes only finitely many elements.

(f) True: it fixes the distinct elements 0 and 1.

(g) True: it fixes the infinite prime subfield $\mathbb{Q}$.

(h) True: composition, identity, and inverse preserve field operations.

(i) True: fixed elements are closed under differences, products and inverses of nonzero elements, and include 0,1.

(j) True. Fixing the larger field E implies fixing F, and these maps form a subgroup.

Exercise 48.30: Synopsis of the existence direction

When α and β have the same minimal polynomial p, identifying both $F(\alpha)$ and $F(\beta)$ with $F[x]/(p)$ gives the unique isomorphism fixing F and sending α to β. This works because evaluation at either root has exactly the kernel (p).

Exercise 48.31: Synopsis of the necessity direction

An isomorphism fixing the coefficients in F carries the equation $p(\alpha) = 0$ to $p(\beta) = 0$, so the minimal polynomials of α and β coincide. Applying the inverse gives the reverse divisibility if one spells this out via minimal polynomials.

Exercise 48.32: Bounding the number of embeddings

An embedding fixing F must send α to a root of its degree-n minimal polynomial. There are at most n such distinct roots in an algebraic closure. Its image on α determines the embedding on every polynomial in α, hence on all $F(\alpha)$. Therefore there are at most n embeddings.

Exercise 48.33: Images of generators determine a map

Every element of $F(\alpha_1, \ldots, \alpha_n)$ is a rational expression in those generators with coefficients in F. A homomorphism fixing F evaluates that expression at $\sigma(\alpha_1), \ldots, \sigma(\alpha_n)$. Its denominator stays nonzero by injectivity. Thus two such automorphisms agreeing on the generators agree on every field element.

Exercise 48.34: Permuting the conjugates present in a field

Let p be the minimal polynomial of α. If $\beta \in E$ satisfies $p(\beta) = 0$, then $p(\sigma(\beta)) = \sigma(p(\beta)) = 0$, because σ fixes every coefficient of p. So σ carries the set of such roots into itself. Its inverse has the same property, proving the restriction is a bijection, hence a permutation.

Exercise 48.35: A generating set has the same fixed field as its group

Since $S \subseteq H$, any element fixed by H is fixed by S. Conversely, if x is fixed by every $s \in S$, it is fixed by s^{-1} too: apply s^{-1} to $s(x) = x$. It is then fixed by any finite composition of generators and their inverses. Those compositions comprise H, so $x \in E_H$. Thus $E_S = E_H$.

Exercise 48.36: Prime cyclotomic fields

(a) A zero ζ of Φ_p is a nonidentity pth root of unity. Since p is prime, it has order p, and $\zeta, \ldots, \zeta^{p-1}$ are distinct, nonidentity roots. They exhaust the $p - 1$ zeros of Φ_p.

(b) Each choice $\sigma_a(\zeta) = \zeta^a$, $1 \leq a < p$, gives a rational embedding onto the same field because a is invertible modulo p. Composition satisfies $\sigma_a \sigma_b = \sigma_{ab}$, proving the group is abelian and has order $p - 1$.

(c) Multiplication by ζ carries the usual basis $1, \ldots, \zeta^{p-2}$ to the basis $\zeta, \ldots, \zeta^{p-1}$. The automorphisms permute its basis vectors transitively. If $\sum_{j=1}^{p-1} c_j \zeta^j$ is fixed by every map, uniqueness of basis coefficients forces all c_j equal, say to c. The sum is then $c(\zeta + \cdots + \zeta^{p-1}) = -c \in \mathbb{Q}$. Conversely rationals are fixed, so the fixed field is $\mathbb{Q}$.

Exercise 48.37: Transcendental generators

Yes. Define $P(\alpha)/Q(\alpha) \mapsto P(\beta)/Q(\beta)$ for $P, Q \in F[x]$, $Q \neq 0$. Transcendence ensures neither evaluation denominator vanishes, and equality of rational expressions is exactly a polynomial identity, so the map is well defined. It preserves both operations and has inverse obtained by exchanging α, β. Thus it is an isomorphism fixing F.

Exercise 48.38: All automorphisms of a rational function field

They are precisely

$$x \longmapsto \frac{ax+b}{cx+d}, \qquad a,b,c,d \in F, \quad ad-bc \neq 0.$$

Such a substitution has inverse $x \mapsto (dx-b)/(-cx+a)$, so it is an automorphism. Multiplying all four coefficients by the same nonzero scalar gives the same map. For completeness, if an automorphism sends x to $r = P(x)/Q(x)$ in lowest terms, put $m = \max(\deg P, \deg Q)$. The standard degree computation gives $[F(x) : F(r)] = m$: $P(T) - rQ(T)$ is the minimal polynomial of x over $F(r)$. Indeed, with r replaced by an independent indeterminate u, the polynomial $P(T) - uQ(T)$ is irreducible in $F[T,u]$, since a factorization of this degree-one polynomial in u would require a nonconstant common divisor of P, Q; Gauss's lemma transfers this to $F(u)[T]$. An automorphism has $F(r) = F(x)$, so $m = 1$. The numerator and denominator are therefore linear or constant, and coprimality/nonconstancy is exactly $ad - bc \neq 0$.

Exercise 48.39: Rigidity of real-field automorphisms

(a) If $u = v^2$, then $\sigma(u) = \sigma(v)^2$. Applying the inverse shows squares map onto squares.

(b) Positive real numbers are exactly the nonzero squares. An automorphism preserves nonzero elements, so it maps positive reals to positive reals.

(c) If $a < b$, then $b - a > 0$, hence $\sigma(b) - \sigma(a) = \sigma(b-a) > 0$. Thus it preserves strict order.

(d) Every automorphism fixes $\mathbb{Q}$. If $\sigma(x) > x$, choose a rational q with $x < q < \sigma(x)$. Order preservation gives $\sigma(x) < \sigma(q) = q$, contradiction. If $\sigma(x) < x$, a rational between them gives the reverse contradiction. Hence $\sigma(x) = x$ for every real x.